Sodium Bicarbonate

Neonatal

FORM

4.2% solution for injection
1ml contains 0.5mmol bicarbonate

4.2% Polyfusor

INDICATION

1. **Neonatal resuscitation within NICU (at consultant discretion)**
2. **For correction of metabolic acidosis.**

USE 4.2% INJECTION

DOSE RANGE

1. DOSE for emergency resuscitation

AGE	DOSE	FREQUENCY	ROUTE
Birth to 6months	2ml /kg of 4.2% injection.	single dose, may be repeated	IV slowly or UVC

2. DOSE for correction of metabolic acidosis (Base deficit)

Step 1 - calculate the infant's bicarbonate deficit

0.3 x BASE DEFICIT (mmol/L) X BODY WEIGHT (kg) = **Full** correction (mmol bicarbonate)

It is usual practice to initially give **half** of this deficit over 1 to 2 hours **then** to reassess base deficit to determine if full correction is needed.

Step 2 - calculate the volume to administer

Sodium bicarbonate 4.2% contains 0.5mmol bicarbonate in each 1ml.

Therefore, give **2ml of 4.2% solution per mmol correction required.**

RECONSTITUTION

Already in solution

DILUTION

Not required

METHOD OF ADMINISTRATION

IV infusion via central line if possible

For correction of base deficit, initially **half of the calculated deficit** should be prescribed and this volume prepared and administered over at least 1-2 hours.

Check glucose, pH and base deficit before administering remaining half of the corrected calculation volume. Sodium Bicarbonate 4.2% injection may be administered undiluted in neonates, but osmolarity is high.

N.B. Sodium bicarbonate is **not** compatible with infusions containing calcium, phosphate or magnesium, including TPN.

It is recommended that where possible, use a separate line and flush well before and after the infusion.

COMPATIBILITY

Solution compatibility	Sodium chloride 0.45%, Sodium chloride 0.9%, Glucose 5%, Glucose 10%
Solution incompatibility	No Information
IV Line compatibility	Fluconazole, furosemide, Heparin, Insulin, metronidazole, Milrinone, Potassium Chloride, vancomycin, digoxin, hydrocortisone, labetalol, morphine, vecuronium, rocuronium
IV Line incompatibility	Amiodarone, adrenaline, calcium gluconate, dobutamine, dopamine, gentamicin, midazolam, noradrenaline, sildenafil. parenteral nutrition solutions, lipid

THIS LIST IS NOT EXHAUSTIVE PLEASE CONTACT PHARMACY FOR FURTHER INFORMATION ON COMPATIBILITY WITH ANY MEDICINES NOT INCLUDED

CAUTIONS, CONTRA-INDICATIONS AND SIDE EFFECTS
- See Summary of Product Characteristics and most recent edition of BNF for Children

FURTHER INFORMATION

- Avoid extravasation. The use of scalp veins should be avoided where possible
- Hypochloraemic alkalosis may occur if sodium bicarbonate is used in conjunction with potassium depleting diuretics such as furosemide. Concurrent use in patients taking potassium supplements may reduce serum potassium concentration by promoting an intracellular ion shift

PH 7-8.5

SODIUM CONTENT 4.2% injection contains 0.5mmol/ml

LICENSED STATUS Sodium bicarbonate 4.2% injection - Unlicensed special
Polyfusor sodium bicarbonate 4.2% infusion 500ml – licensed

APPLICABLE POLICIES West of Scotland Neonatal Guidelines

Consult local policy if applicable

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Administer reconstituted solutions immediately.
All vials, ampoules and infusion bags are for single use only unless otherwise stated.

Dose may vary depending on indication, age, renal function, hepatic function, and concomitant medications.
This monograph should be used in conjunction with the terms of reference document prepared by the West of Scotland Pharmacist Group. Information is correct at the time of publication and as per local practice agreement.
For further advice please contact your clinical pharmacist or pharmacy department